

Iron-Cycling CW-MFC Enhanced Removal of Sartans

Zhang et al. | Environmental Science & Technology | 2026

Abstract

This study investigates the removal of pharmaceutical contaminants (sartans) using iron-cycling constructed wetland microbial fuel cells (CW-MFC). Results show a 94.3% removal efficiency under optimized conditions. The anodic iron oxidation drives electron transfer, enhancing microbial degradation pathways.

1. Introduction

Emerging contaminants such as angiotensin receptor blockers (ARBs) pose significant risks to aquatic ecosystems. Conventional wastewater treatment plants fail to fully remove these compounds. Constructed wetlands coupled with microbial fuel cells represent a promising low-energy alternative.

2. Materials and Methods

CW-MFC reactors (30 L) were packed with iron filings and gravel media. Synthetic wastewater spiked with valsartan (10 mg/L) was fed at HRT = 3 days.

3. Results

Table 1 summarizes contaminant removal across experimental runs. At Fe(II) concentration of 5 mM, sartan removal reached 94.3 +/- 1.2%. Voltage output stabilized at 320 mV after 14 days of operation.

4. Discussion

The enhanced removal is attributed to Fenton-like reactions generated by Fe(II)/Fe(III) cycling. The MFC component accelerates iron oxidation at the anode, sustaining reactive oxygen species production. Microbial community analysis revealed enrichment of *Geobacter* and *Desulfovibrio* species.

5. Conclusion

Iron-cycling CW-MFC systems achieve high sartan removal with concurrent electricity generation. This technology offers a sustainable approach to pharmaceutical wastewater treatment.